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NEAR AND FAR-FIELD DEFORMATION FROM THE 2023 TURKEY EARTHQUAKES USING GNSS DATA.

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EUROPEAN GEOSCIENCES UNION GENERAL ASSEMBLY 2025 VIENNA | AUSTRIA | 27 APR. - 02 MAY 2025



INTRODUCTION

On February 6th, 2023 at 01:17:35 UTC the M_w 7.8 Nurdagi-Pazarcik earthquake nucleated ~ 15 km southeast of the mapped trace of the East Anatolian Fault Zone. Relocations place the hypocenter at (37.0234° E, 37.2444° N, depth=12 km) and analyses of teleseismic data show a left-lateral source mechanism on a vertical or near vertical fault. A vigorous aftershock sequence followed and a little over 9 hours after the first event, at 10:24:49 UTC, the M_w 7.6 [?]

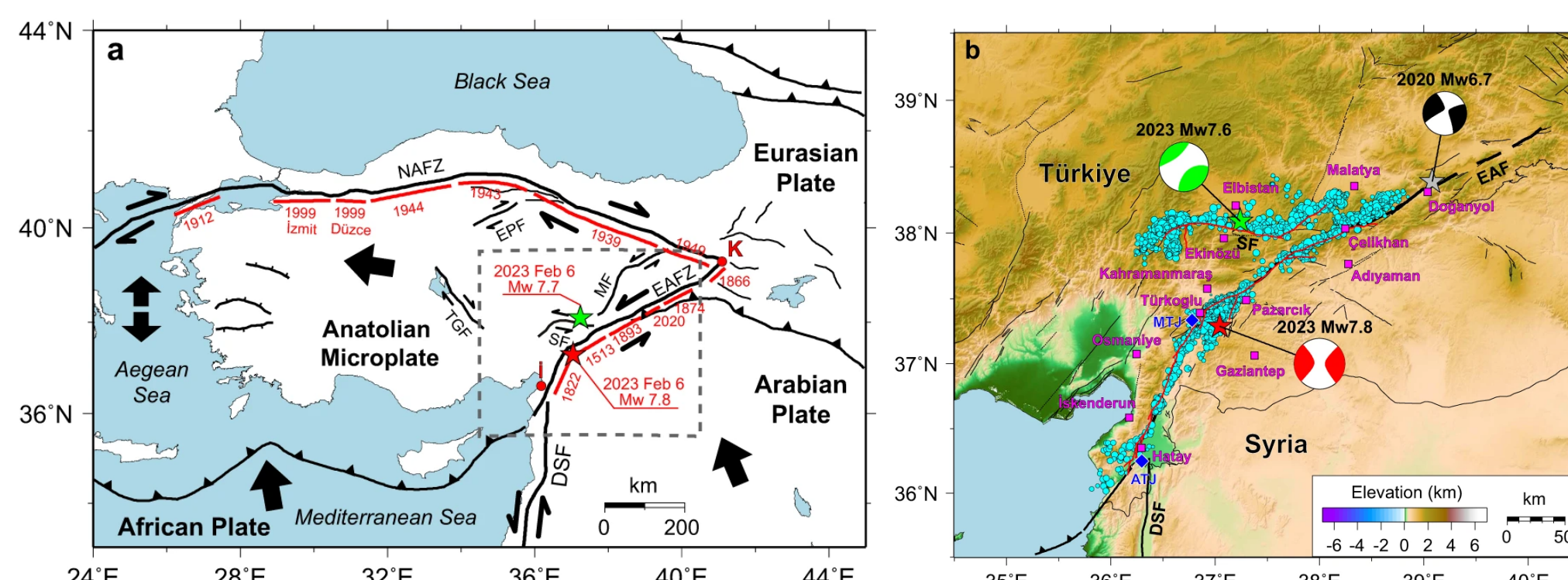


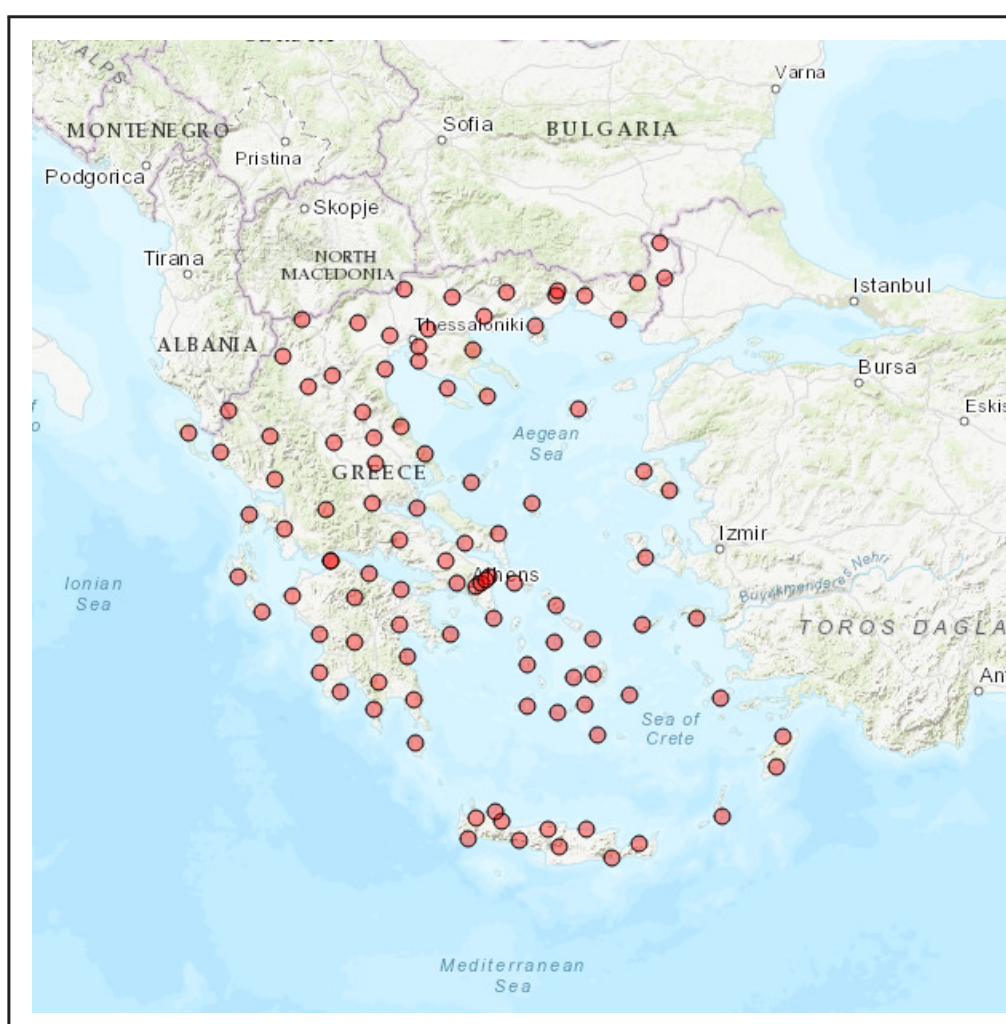
Fig.1: Tectonic background (a) and focal mechanisms (b) of 2023 Turkey earthquake doublet

Abstract Number: EGU2025-18416



HEPOS NETWORK

Hellenic Cadastre has since 2007 established a network of 98 continuous GPS sites, spatially covering the whole country. The network is since maintained by the office, and its products and data used for surveying and supporting national infrastructure.



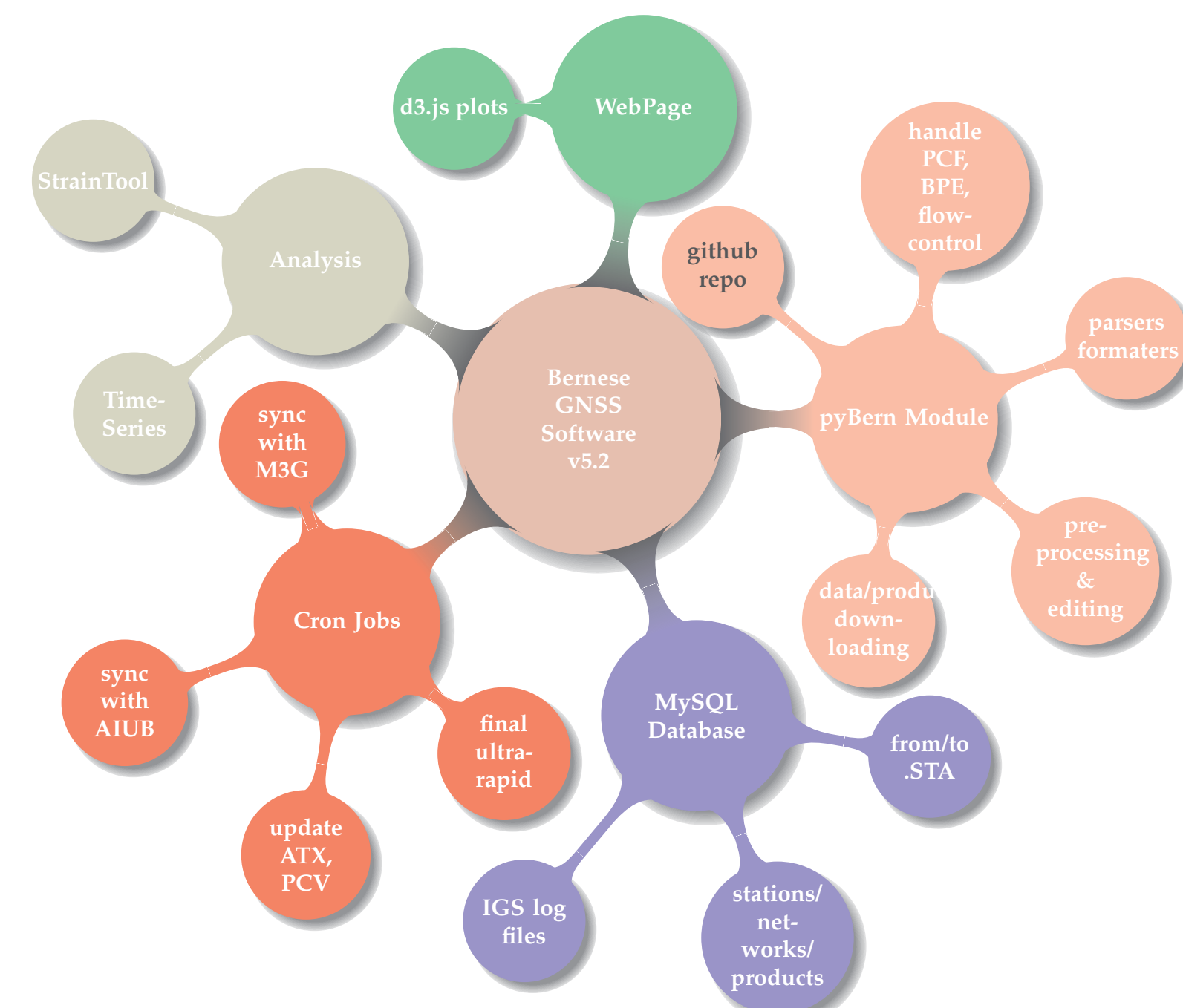
In 2020, Hellenic Cadastre successfully concluded a large-scale upgrade of its network, introducing a full switch to multi-GNSS. All network sites were equipped with modernized firmware and hardware to support all major GNSS systems (GPS, Glonass, Galileo and Beidou).

For this study we analyzed daily RINEX (v.2) files of four individual years, namely 2011, 2015, 2021 and 2022. The data were provided under a special agreement from Hellenic Cadastre to

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DATA ANALYSIS

The data described above were analyzed using the Bernese GNSS Software (v. 5.2) (Dach et al., 2015). Both GPS and GLONASS observation were used when available (most of 2021 and 2022). IGB14 was used as the frame of reference, using a set of 19 IGS stations for alignment. CODE final products were incorporated in the analysis, consistent with the frame of choice. The key parameter of interest was station coordinates.



Processing is consistent with EUREF standards (Guidelines for Analysis Centres).

- SINEX with required info/blocks,
- Reference frame **IGb14**,
- IERS Conventions 2010,
- IGS/CODE products,
- ocean loading corrections (**FES2004**),
- 3° elevation cut-off angle; elevation dependent weighting,
- GMF and/or **VMF1**; Chen-Herring gradient parameter,
- ambiguities fixed (length-dependent algorithm),
- use GLONASS obs (when available)
- use ATX files (**eprn_14.atx**) - **individual calibrations**

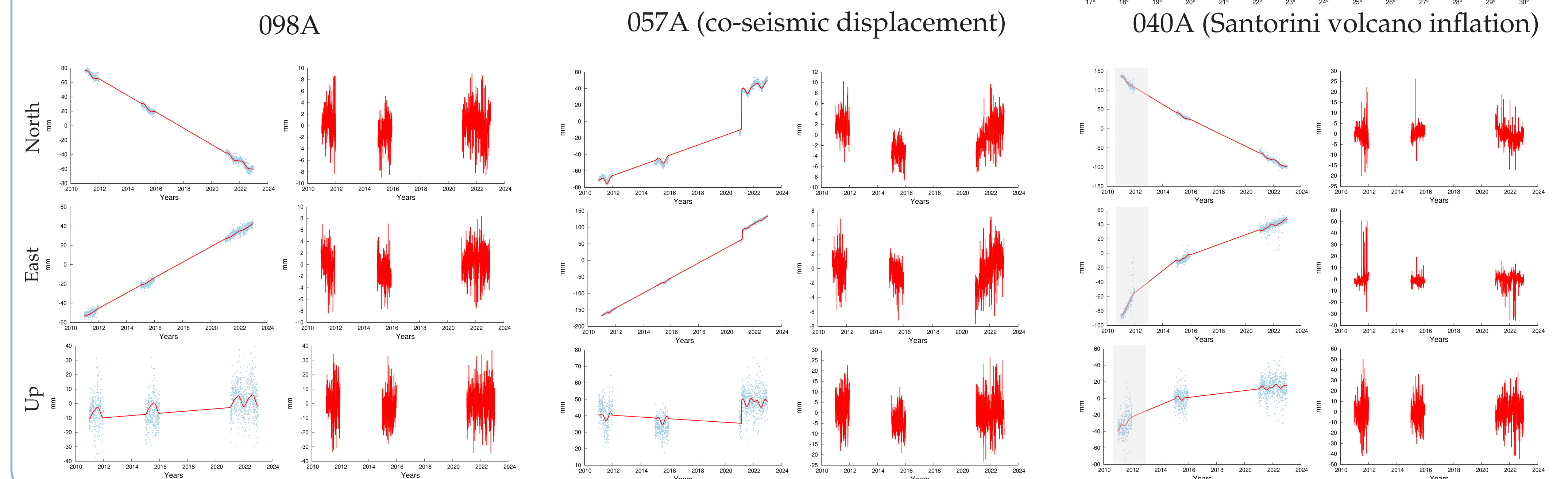
ESTIMATION OF VELOCITY FIELD

Following the estimation of daily site coordinates, position time-series were stacked for every site of the network. The latter were in turn analyzed using the Hector software (Bos et al., 2013) for the estimation of tectonic velocities and harmonic signals. The time span of more than ten years allows estimation of site velocities, yet the temporal density of data could inhibit the accuracy that can be obtained.

Time series analysis of three different stations are presented. Velocity field with respect to IGB14 presented on the following map.

Velocities vary per component:

comp	min (mm/yr)	max (mm/yr)
North	-17.9	15.5
East	2.4	26.0
Up	-5.5	4.7

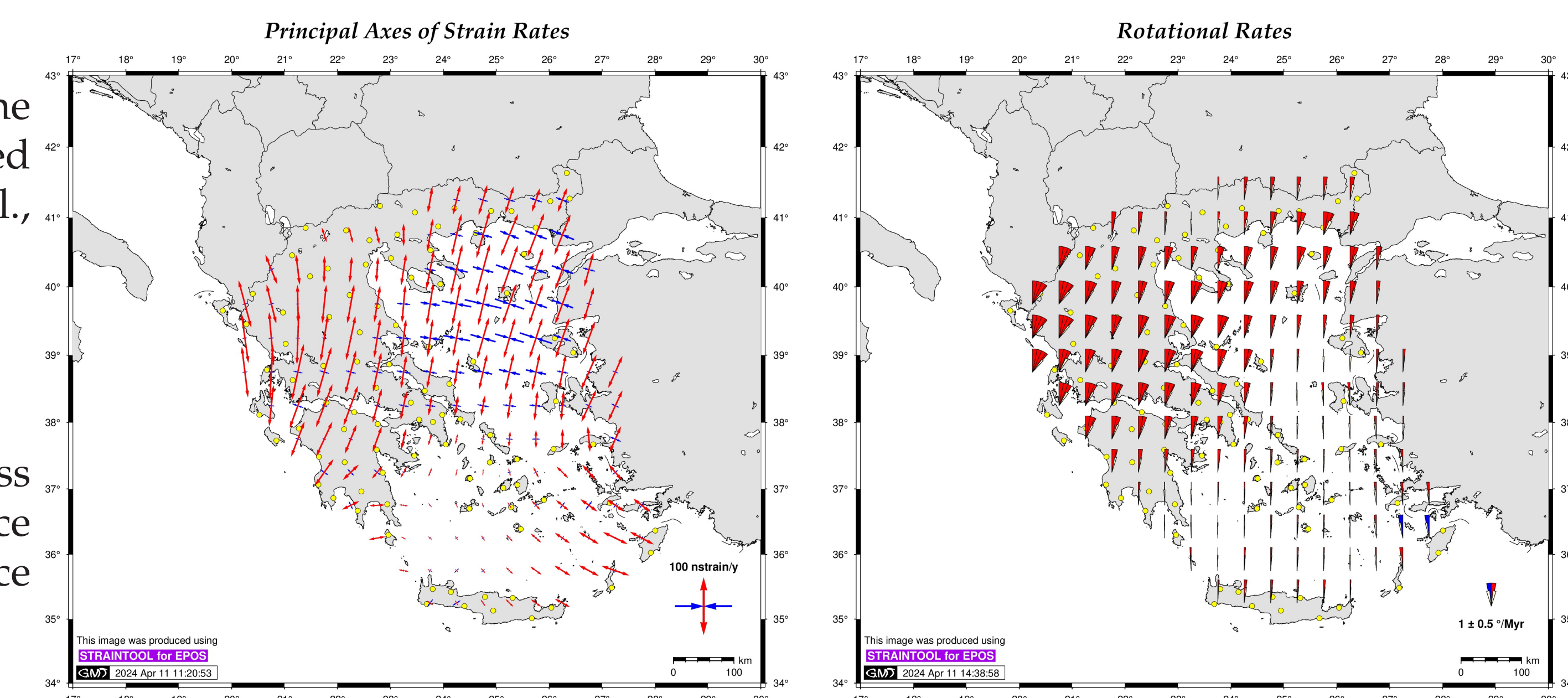


STRAIN AND ROTATIONAL RATES

Strain and rotational rates presented in the following maps. We present results obtained using StrainTool Software (Anastasiou et al., 2021) and the following input parameters:

grid-size	ltype	Wt
0.5°x0.5°	gaussian	24

Overall, our first results reproduce the gross features of tectonic deformation in Greece such as N-S Extension in mainland Greece and clockwise rotational rates on the west.



FUTURE RESEARCH

We consider this study to be a work-in-progress; we plan to go on with data analysis for all the available years. Currently, we are working on :

- the analysis of vertical velocities
- the investigation of the deformation field in smaller areas

FUNDING BY HELLENIC CADASTRE

We acknowledge support for this research by the project "Support of Permanent GNSS station Network of Hellenic Cadastre" funded by the **Hellenic Cadastre**.

Authors want to thank you Hellenic Cadastre providing the data of HEPOS Network.

